IS221: Web Applications Development

Faculty of Science, Technology and Environment
School of Computing, Information and Mathematical Sciences

Final Examination-Semester I 2018

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 180 minutes

INSTRUCTIONS:

1. This examination has four sections:
   a. Section A: Multiple Choice (20 marks)
   b. Section B: Coding (34 marks)
   c. Section C: Short and Long Answers (46 marks)
2. **All** questions are compulsory
3. Write **ALL** answers in the spaces provided within this paper.
4. This exam is **marked out of 100** and is worth **50%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam** which is equivalent to **40 marks out of 100**
6. This is a closed book exam.
7. This paper has 24 pages including this page.

Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question in the provided answer sheet on page 7. There is only **one** possible answer.

1) To center a navigation bar, set the ____ attribute to "center" within the ____ tag.

   a. align;
   b. ; align
   c. align;
   d. title;<body>

2) Which of the following has the least weight in the style sheet order of precedence?

   a. Author's style sheet
   b. W3C's standard style sheet
   c. User's style sheet
   d. Browser's default style sheet

3) When creating an internal link, one of the <a> anchor elements targets the fragment identifier name in the ____ attribute.

   a. name
   b. color
   c. href
   d. destination

4) If the border-width property has two values, the ____ border width(s) is/are set to the first value, and the ____ border width(s) is/are set to the second value.

   a. top and bottom; right and left
   b. right and left; top and bottom
   c. top and right; bottom and left
   d. bottom and right; top and left

5) Using the example “p {border-style: solid double dashed dotted ;}”, the bottom border of a <p> element will be ____.

   a. solid
   b. double
   c. dashed
   d. dotted

6) You should keep a ____ hierarchy in order to create manageable information segments.

   a. circular
   b. flat
   c. three-dimensional
   d. multi-tiered

7) CSS comments begin with ____.

   a. `/?`
   b. `/*`
   c. `*/`
   d. `?/`

8) ____ elements appear as blocks such as paragraphs.

   a. List-item
   b. Page-level
   c. Inline-level
   d. Block-level

9) ____ values are always relative to another value

   a. Point
   b. Absolute
   c. Percentage
   d. Pixel

10) The default color for an active link is ____.

a. purple
b. blue
c. green
d. red

11) In CSS, the ____ is equal to the font size of an element.

a. em
b. pt
c. pc
d. ex

12) To make sure that the frame border and spacing are turned off for all browsers, use the border attribute set to "____".

a. 0
b. none
c. blank
d. hidden

13) The ____ type input object adds a graphic button to the form, rather than the default button.

a. picture
b. image
c. img
d. button

14) When limiting information overload, you can judge your page length by pressing the Page Down key; if you have to press it more than ____ times to move from the top to bottom of your page, break up the content.

a. one or two
b. two or three
c. three or four
d. four or five

15) The following style rule selects an <a> element: ____.

a. {a font-size: 85%; font-weight: bold; font-family: arial, sans-serif;}
b. a font-size: 85%; font-weight: bold; font-family: arial, sans-serif;
c. a{font-size: 85%; font-weight: bold; font-family: arial, sans-serif;}
d. a{font-size: 85% font-weight: bold font-family: arial, sans-serif}

16) When using a Master Page for the layout of a web page, the _____ controls identify locations where content may be inserted in a content page.

a. ContentPlaceHolder
b. FormContentHolder
c. PagePlaceHolder
d. DivisionPlaceHolder

17) ____ is/are especially effective in guiding users who may have arrived at a page from somewhere outside this Web site:

a. A site map at the bottom of the page
b. A linked path at the top of the page showing the user's location within the site hierarchy
c. Linked graphics on the right side of the page
d. A glossary link

18) When using flexible layout for a web page, setting the minimum and maximum widths enables control of _____ and _____ of the web page content Dynamic

a. dilation, expansion
b. pixilation, expansion
c. compression, expansion
d. pixilation, compression

19) a:link {color: #0000FF;} will result in:

   a. Visited links are blue
   b. Active links are blue
   c. New links are blue
   d. All of the above

20) Given the style declarations below and the tag <h1> Welcome </h1>, Welcome will be colored _____
   External Stylesheet – h1 {color: blue ;}
   Internal Stylesheet - h1 {color: green ;}
   Inline style - h1 {color: yellow ;}

   a. colorless
   b. green
   c. blue
   d. yellow

Section A: Answer sheet

Circle the correct answer.

1.	**a**	**b**	**c**	**d**
2.	**a**	**b**	**c**	**d**
3.	**a**	**b**	**c**	**d**
4.	**a**	**b**	**c**	**d**
5.	**a**	**b**	**c**	**d**
6.	**a**	**b**	**c**	**d**
7.	**a**	**b**	**c**	**d**
8.	**a**	**b**	**c**	**d**
9.	**a**	**b**	**c**	**d**
10.	**a**	**b**	**c**	**d**
11.	**a**	**b**	**c**	**d**
12.	**a**	**b**	**c**	**d**
13.	**a**	**b**	**c**	**d**
14.	**a**	**b**	**c**	**d**
15.	**a**	**b**	**c**	**d**
16.	**a**	**b**	**c**	**d**
17.	**a**	**b**	**c**	**d**
18.	**a**	**b**	**c**	**d**
19.	**a**	**b**	**c**	**d**
20.	**a**	**b**	**c**	**d**

Section B: Coding Questions 34 Marks

You have been employed to revamp the Rental Advertisement website. Currently rental posting is a paper-based process that needs to be made available online for the public.

Question 1

Given the code for the Rental Advertisement posting page depicted in Listing 1 on pages 9, 10 and 11, draw the web form when it is rendered. **(10 marks)**

Listing 1 – Web page code

```
<%@ Page Title=" Rental Advertisement " Language="VB" MasterPageFile="~/Site.Master"
AutoEventWireup="false" CodeFile="Posting.aspx.vb" Inherits="_Default" %>
<asp:Content ID="HeaderContent" runat="server" ContentPlaceHolderID="HeadContent">
    <style type="text/css">
       .style1
       {
      color: black;
      text-align: center;
      font-family: FangSong;
       }
       .style2
       {
          width: 600px;
       }
       .style3
       {
          height: 38px;
       }
       .style5
       {
          font-family: FangSong;
          color: black;
       }
       .style6
       {
      text-align: center;
      color: red;
      font-family: "AR CHRISTY";
       }
       .style7
       {
          color: green;
       }
       .style8
       {
          color: blue;
       }
    </style>
</asp:Content>
<asp:Content ID="BodyContent" runat="server" ContentPlaceHolderID="MainContent">
<h1 class="style6">
        <strong> Rental Advertisement posting</strong></h1>
<p class="style1">
  <asp:Image ID="Image1" runat="server" ImageUrl="~/Images/RA_logo.PNG" Height="200px"
    Width="100%" />
</p>
<p class="style1"> </p>
<table class="style2" align="center">
```

```
 <tr>
     <td> First Name:</td>
     <td>
     <asp:TextBox ID="txtFirstName" runat="server" Width="160px"></asp:TextBox>
     </td>
 </tr>
 <tr>
     <td> Last Name:</td>
     <td>
     <asp:TextBox ID="txtLastName" runat="server" Width="160px"></asp:TextBox>
     </td>
 </tr>
 <tr>
     <td> Mobile Number:</td>
     <td>
     <asp:TextBox ID="txtMobileNumber" runat="server" Width="156px" TextMode="Phone">
     </asp:TextBox>
     </td>
</tr>
<tr>
     <td>
          <h2 class="style5"><span class="style7"> Property Type </span>:</h2>
     </td>
     <td>
     <asp:DropDownList ID="ddlPropertyType" runat="server">
          <asp:ListItem>Flat</asp:ListItem>
          <asp:ListItem>House</asp:ListItem>
          <asp:ListItem>Apartment</asp:ListItem>
          <asp:ListItem>Others</asp:ListItem>
     </asp:DropDownList>
     <asp:XmlDataSource ID="xdsProperties" runat="server"
             DataFile="~/App_Data/Properties.xml">
     </asp:XmlDataSource>
     </td>
 </tr>
 <tr>
     <td> Rental Price:</td>
     <td>
         <asp:TextBox ID="txtPrice" runat="server" Width="156px" TextMode="Number">
         </asp:TextBox>
     </td>
</tr>
<tr>
     <td> Furnished:</td>
     <td>
         <asp:RadioButtonList ID="rblFurnished " runat="server"
              RepeatDirection="horizontal" TextAlign="Left">
                  <asp:ListItem>Yes</asp:ListItem>
                  <asp:ListItem>No</asp:ListItem>
         </asp:RadioButtonList>
     </td>
```

```
  </tr>
  <tr>
      <td>
      <h3 class="style5"><span class="style8"> Amenities: </span>:</h3>
      </td>
          <td>
                <asp:CheckBoxList ID="cblSubscriptions" runat="server" RepeatColumns="2">
                    <asp:ListItem>Garage</asp:ListItem>
                    <asp:ListItem>Balcony</asp:ListItem>
                    <asp:ListItem>Burglar Bars</asp:ListItem>
                    <asp:ListItem>Swimming Pool</asp:ListItem>
                    <asp:ListItem>Fenced</asp:ListItem>
                    <asp:ListItem>Attic</asp:ListItem>
                </asp:CheckBoxList>
           </td>
  </tr>
  <tr>
           <td class="style3"> </td>
           <td class="style3"> <br />
               <asp:Button ID="btnSubmit" runat="server" Text="Submit"
                    style="text-align: center" ForeColor="red" />
           </td>
  </tr>
 </table>
</asp:Content>
```

Question 2

Analyze the use of CSS in the Style section on the code provided in Listing1 on pages 9, 10 and 11. Write a style that will have all headings to be colored maroon, left aligned, font to be Verdana, with background color of yellow, and all paragraphs to be colored blue. **(5 marks)**

Question 3

a) A Page Directive appears on top of each page or web form as in Listing1 on page 9. What is the main purpose of the Page Directive? **(1 mark)**

b) What are the names of the web form and the corresponding code behind the page file in Listing1 on page 9 **(2 marks)**

c) Page directive of a web form can determine if the page is a normal web form or a master page. What is the main difference, although there are other differences as well?

(2 marks)

Question 4

In Listing 1 – on page 10, a DropDownList server control or object is used. This was named as ddlPropertyType, which simply gives a list of possible property type. Write the corresponding XHTML code for the ddlPropertyType DropDownList server control.

(4 marks)

Question 5

Figure 1 below shows the body structure of the Register.aspx web form, whereas Figure 2 shows the body structure of the corresponding XHTML file generated and sent by the ASP.NET application from the server to the client's browser. Both have a <form> element, which contains all codes.

```
 9  <body>
10      <form id="form1" runat="server">
11      <div>...</div>
79      </form>
80  </body>
```

Figure 1 Web Form body

```
 9  <body>
10      <form
11      <div class="aspNetHidden">...</div>
14      <div class="aspNetHidden">...</div>
19      <div>...</div>
86      </form>
87  </body>
```

1 2 3

Figure 2 Web Form code as rendered in the client's browser.

a) Note that in Figure 2, the XHTML file, line 10, the **<form>** element's three (3) attributes are hidden and labelled 1, 2, and 3. Name these three hidden attributes and their corresponding values.

(6 marks)

Label	Value with attributes
1	
2	
3	

b) A view of the Web page source code in Figure 2 shows that two hidden input objects were added to the page as a result of the processing on the server. Name these two hidden input objects. **(2 marks)**

c) The Register.aspx Web Form also used a Label server control or object to display the information the user entered. This was named as lblVerify and the ASP.NET code is:

```
<asp:Label ID="lblVerify" runat="server"></asp:Label>
```

Write the corresponding XHTML code for this Label server control. **(2 marks)**

Section C: Short/Long Answer Questions 46 Marks

Question 1

Using a diagram explain CSS box model. **(5 marks)**

Question 2

Effective navigation includes providing not only links to other pages in the Web site, but also cues to user's location. List the four navigation questions that users should be able to answer when navigating through your Web site. For each of the questions: **(10 marks)**

- Identify the type of information that the website is required to provide
- Identify tools/techniques that a website could use to provide this information

Student ID#:_____ Student Name: _____

Navigation Question 1: where am i

Type of information:

Tool/technique:
breadcrumb

Navigation Question 2: where can i go?

Type of information:

Tool/technique:
related links

Navigation Question 3: how do i get there?

Type of information:

Tool/technique:
site map or contextual links

Navigation Question 4: how do i get back to where i started?

Type of information:

Tool/technique:
page turners and or linked navigation

Question 3

Using a diagram, illustrate and describe ASP.net application processing. **(10 marks)**

Question 4

You have been employed to re-design Rental Advertisement's website and are required to integrate the following site requirements:

- Logo and mission statement should be featured
- Search box
- Links to the other pages of the website should be available
- Rental property specials should be a main feature content
- Advertisements of rental related information, deals and news should also feature
- Copyright statement should feature as well
- Links to all other pages should be available on all pages
- Uniform look and feel for all the pages, the only difference should be the actual content

a) With the aid of a diagram, illustrate how the Home page template would look like

(5 marks)

b) With the aid of a diagram, illustrate how you would design a responsive version of the homepage for mobile devices that have screen sizes ranging from 320px to 768px.

(5 marks)

Question 5

Discuss how you would design for both Accessibility and Legibility **(3 Marks)**

Question 6

Discuss how you are able to view your Facebook page on your mobile phone. Your discussion should include the core elements of web applications architecture. **(3 Marks)**

Question 7

How do you specify overall and individual padding properties? Explain with examples **(2 Marks)**

Question 8

Describe the processes of removing the hypertext border from an image, aligning text and images, and adding white space around images. **(3 Marks)**

Web form Draft Page

Web form Draft Page

THE END